

# PADHAISHURU

GATE Preparation Platform

## GATE EY

Ecology & Evolution

### TREND-BASED MOCK PAPER

— 03 —

|                 |             |
|-----------------|-------------|
| Duration        | 3 Hours     |
| Total Questions | 65          |
| Maximum Marks   | 100         |
| Sections        | 10 subjects |

#### Difficulty Distribution

Easy: 20 questions (31%)

Moderate: 33 questions (51%)

Difficult: 12 questions (18%)

Unofficial practice material. Not affiliated with or endorsed by GATE/IIT.  
These mock papers are constructed using analysis of historical GATE  
PYQ patterns.  
Historical patterns do not guarantee future examination questions.

# GENERAL INSTRUCTIONS

GATE EY — Trend-Based Mock Paper

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## 1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

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## 2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
  - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
  - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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## 3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
  - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
  - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
  - NAT questions carry no negative marking.
  - MSQ questions carry no negative marking.
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## 4. How to Answer

- For MCQ: Select exactly one option.
  - For MSQ: Select one or more options. Partial credit is NOT awarded.
  - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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## 5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

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## 6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

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## 7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
  - The Question Booklet contains multiple pages. Verify the page count.
  - Immediately report any discrepancy in the Question Booklet to the invigilator.
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**Q1.** 1M [-0.33]

*[General Aptitude]*

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

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**Q2.** 1M [-0.33]

*[Engineering Mathematics]*

Standard deviation is:

- (A) Mean
- (B) Spread
- (C) Median
- (D) Mode

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**Q3.** 1M [-0.33]

*[Ecology]*

Logistic growth has:

- (A) No limit
- (B) Carrying capacity
- (C) Exponential
- (D) None

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**Q4.** 1M [-0.33]

*[Evolution]*

Darwin finches from:

- (A) Galapagos
- (B) Africa
- (C) Asia
- (D) Australia

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**Q5.** 1M [no -ve]

*[Genetics]*

Hardy-Weinberg needs:

\_\_\_\_\_

*Enter your answer here*

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**Q6.** 1M [-0.33]

*[Environmental Science]*

Biodiversity hotspot criteria:

- (A) Endemism
  - (B) Deforestation
  - (C) Both
  - (D) None
-

**Q7.**

1M [-0.33]

*[Cell Biology]*

Mitochondria function:

- (A) Protein synthesis
- (B) ATP production
- (C) Photosynthesis
- (D) None

**Q8.**

1M [-0.33]

*[Plant Physiology]*

Photosynthesis in:

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Ribosome

**Q9.**

1M [-0.33]

*[Zoology]*

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

**Q10.**

1M [no -ve]

*[Behavioral Ecology]*

Monarch butterfly migrates from:

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*Enter your answer here***Q11.**

1M [-0.33]

*[Behavioral Ecology]*

Monarch butterfly migrates from:

- (A) Canada to Mexico
- (B) India
- (C) Africa
- (D) Europe

**Q12.**

1M [no -ve]

*[Behavioral Ecology]*

Monarch butterfly migrates from:

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*Enter your answer here***Q13.**

1M [-0.33]

*[Ecology]*

Logistic growth has:

- (A) No limit
- (B) Carrying capacity
- (C) Exponential
- (D) None

**Q14.**

1M [-0.33]

*[Environmental Science]*

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

**Q15.**

1M [-0.33]

*[Zoology]*

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

**Q16.**

1M [no -ve]

*[Plant Physiology]*

Photosynthesis in:

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*Enter your answer here***Q17.**

1M [no -ve]

*[Genetics]*

Hardy-Weinberg needs:

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*Enter your answer here***Q18.**

1M [no -ve]

*[Plant Physiology]*

Photosynthesis in:

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*Enter your answer here***Q19.**

1M [no -ve]

*[Behavioral Ecology]*

Monarch butterfly migrates from:

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*Enter your answer here***Q20.**

1M [-0.33]

*[Evolution]*

Darwin finches from:

- (A) Galapagos
- (B) Africa
- (C) Asia
- (D) Australia

**Q21.**

1M [-0.33]

*[Environmental Science]*

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

**Q22.**

1M [-0.33]

*[Genetics]*

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

**Q23.**

1M [-0.33]

*[Cell Biology]*

Mitochondria function:

- (A) Protein synthesis
- (B) ATP production
- (C) Photosynthesis
- (D) None

**Q24.**

1M [-0.33]

*[Ecology]*

Logistic growth has:

- (A) No limit
- (B) Carrying capacity
- (C) Exponential
- (D) None

**Q25.**

1M [-0.33]

*[General Aptitude]*

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

**Q26.**

1M [-0.33]

*[Cell Biology]*

Mitochondria function:

- (A) Protein synthesis
- (B) ATP production
- (C) Photosynthesis
- (D) None

**Q27.**

1M [-0.33]

*[Plant Physiology]*

Photosynthesis in:

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Ribosome

**Q28.**

1M [no -ve]

*[Engineering Mathematics]*

Standard deviation is:

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*Enter your answer here***Q29.**

1M [-0.33]

*[Environmental Science]*

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

**Q30.**

1M [-0.33]

*[Zoology]*

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

**Q31.**

2M [-0.66]

*[Zoology]*

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

**Q32.**

2M [no -ve]

*[Cell Biology]*

Mitochondria function:

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*Enter your answer here***Q33.**

2M [no -ve]

*[Ecology]*

Logistic growth has:

---

*Enter your answer here*

**Q34.**

2M [no -ve]

*[Engineering Mathematics]*

Standard deviation is:

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*Enter your answer here***Q35.**

2M [-0.66]

*[Evolution]*

Darwin finches from:

- (A) Galapagos
- (B) Africa
- (C) Asia
- (D) Australia

**Q36.**

2M [no -ve]

*[Behavioral Ecology]*

Monarch butterfly migrates from:

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*Enter your answer here***Q37.**

2M [no -ve]

*[General Aptitude]*

Antonym of 'indigenous':

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*Enter your answer here***Q38.**

2M [-0.66]

*[Genetics]*

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

**Q39.**

2M [no -ve]

*[Zoology]*

Binomial nomenclature by:

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*Enter your answer here***Q40.**

2M [no -ve]

*[Behavioral Ecology]*

Monarch butterfly migrates from:

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*Enter your answer here***Q41.**

2M [-0.66]

*[Zoology]*

Binomial nomenclature by:

- (A) Linnaeus
- (B) Darwin
- (C) Mendel
- (D) Watson

**Q42.**

2M [-0.66]

*[Environmental Science]*

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

**Q43.**

2M [-0.66]

*[General Aptitude]*

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

**Q44.**

2M [no -ve]

*[Plant Physiology]*

Photosynthesis in:

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*Enter your answer here***Q45.**

2M [-0.66]

*[Plant Physiology]*

Photosynthesis in:

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Ribosome

**Q46.**

2M [no -ve]

*[Evolution]*

Darwin finches from:

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*Enter your answer here***Q47.**

2M [-0.66]

*[Genetics]*

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

**Q48.**

2M [-0.66]

*[Genetics]*

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

**Q49.**

2M [no -ve]

*[Ecology]*

Logistic growth has:

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*Enter your answer here***Q50.**

2M [no -ve]

*[Evolution]*

Darwin finches from:

- (A) Galapagos
- (B) Africa
- (C) Asia
- (D) Australia

**Q51.**

2M [-0.66]

*[Genetics]*

Hardy-Weinberg needs:

- (A) No mutation
- (B) No migration
- (C) Large pop
- (D) All

**Q52.**

2M [no -ve]

*[Environmental Science]*

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

**Q53.**

2M [-0.66]

*[Cell Biology]*

Mitochondria function:

- (A) Protein synthesis
- (B) ATP production
- (C) Photosynthesis
- (D) None

**Q54.**

2M [no -ve]

*[Environmental Science]*

Biodiversity hotspot criteria:

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*Enter your answer here***Q55.**

2M [no -ve]

*[Behavioral Ecology]*

Monarch butterfly migrates from:

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*Enter your answer here*

**Q56.**

2M [-0.66]

*[Plant Physiology]*

Photosynthesis in:

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Ribosome

**Q57.**

2M [-0.66]

*[Environmental Science]*

Biodiversity hotspot criteria:

- (A) Endemism
- (B) Deforestation
- (C) Both
- (D) None

**Q58.**

2M [-0.66]

*[General Aptitude]*

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

**Q59.**

2M [-0.66]

*[Plant Physiology]*

Photosynthesis in:

- (A) Mitochondria
- (B) Chloroplast
- (C) Nucleus
- (D) Ribosome

**Q60.**

2M [-0.66]

*[Engineering Mathematics]*

Standard deviation is:

- (A) Mean
- (B) Spread
- (C) Median
- (D) Mode

**Q61.**

2M [no -ve]

*[General Aptitude]*

Antonym of 'indigenous':

- (A) Native
- (B) Foreign
- (C) Local
- (D) Original

**Q62.**

2M [-0.66]

*[Cell Biology]*

Mitochondria function:

- (A) Protein synthesis
- (B) ATP production
- (C) Photosynthesis
- (D) None

**Q63.**

2M [no -ve]

*[Evolution]*

Darwin finches from:

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*Enter your answer here***Q64.**

2M [no -ve]

*[Engineering Mathematics]*

Standard deviation is:

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*Enter your answer here***Q65.**

2M [-0.66]

*[Engineering Mathematics]*

Standard deviation is:

- (A) Mean
- (B) Spread
- (C) Median
- (D) Mode